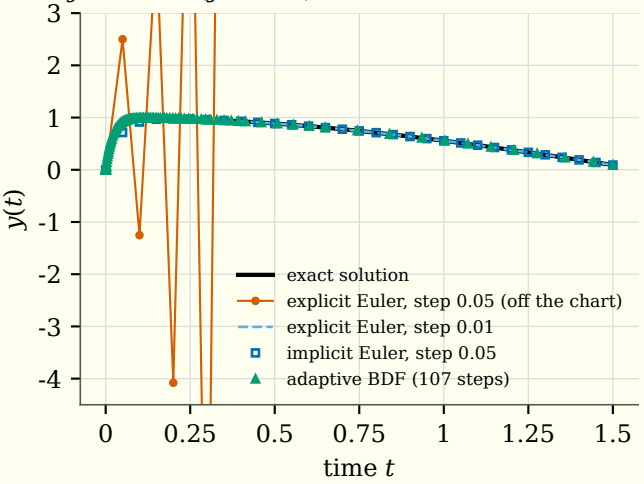
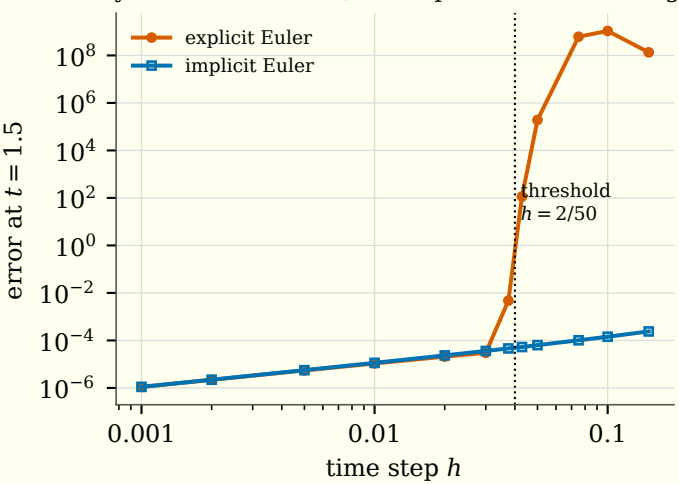

dy/dt = −50 (y − cos t) : the naive method blows up
y(t)
time t
exact solution
explicit Euler, step 0.05 (off the chart)
explicit Euler, step 0.01
implicit Euler, step 0.05
adaptive BDF (107 steps)


Beyond the threshold, the explicit method diverges
explicit Euler
implicit Euler
error at t = 1.5
threshold
h = 2/50
time step h